

CSD-5004 Mathematical Foundations for Data Science Lab

Assignment 1: Python Fundamentals for Data Science

Numerical Computation and Precision

1. Accept a positive number and compute its square root.
2. Display the result rounded to 2, 5, 10, and 15 decimal places.
3. Investigate floating-point arithmetic by evaluating expressions involving decimal numbers and explain your observations.
4. Determine the machine epsilon for floating-point numbers using Python.

Vector Operations

1. Accept two vectors of equal length from the user.
2. Perform vector addition and subtraction.
3. Compute element-wise multiplication.
4. Compute the dot product.
5. Calculate the Euclidean norm of each vector.
6. Compute the cosine similarity between the vectors.

Matrix Operations

1. Accept the dimensions and elements of a matrix.
2. Compute its transpose.
3. If the matrix is square:
 - Compute its determinant.
 - Find its inverse (if it exists).
 - Compute its eigenvalues and eigenvectors.
4. Verify the inverse by matrix multiplication.

Probability Through Simulation

1. Accept the number of trials from the user.
2. Simulate the experiment of rolling two dice repeatedly.
3. Estimate the probability of obtaining a user-specified sum.
4. Compare the simulated probability with the theoretical probability.
5. Plot the frequency distribution of all possible sums.

Function Analysis and Visualization

1. Accept the coefficients of a polynomial function from the user.
2. Plot the function over a suitable range.

3. Estimate the derivative numerically.
4. Plot the derivative on the same graph.
5. Identify approximate stationary points from the graph.

Statistical Analysis of Data

1. Accept a list of numerical values from the user.
2. Compute:
 - Mean
 - Median
 - Mode
 - Variance
 - Standard deviation

Descriptive Statistics Without Libraries

1. Accept a dataset from the user.
2. Write your own functions to compute:
 - Mean
 - Median
 - Mode
 - Range
 - Standard deviation
3. Compare the results with those obtained using NumPy or Pandas.

Linear Algebra Concepts

1. Accept a matrix from the user.
2. Determine its rank.
3. Check whether its columns are linearly independent.
4. Explain the implications of linear dependence in a machine learning dataset.
5. If the matrix is square, compute its condition number and discuss the result.

Mathematical Problem Solving Using Python

1. Accept an integer nnn .
2. Compute:
 - Sum of first nnn natural numbers
 - Sum of squares of first nnn natural numbers
 - Sum of cubes of first nnn natural numbers